PM2.5 LEVELS AND ASTHMA

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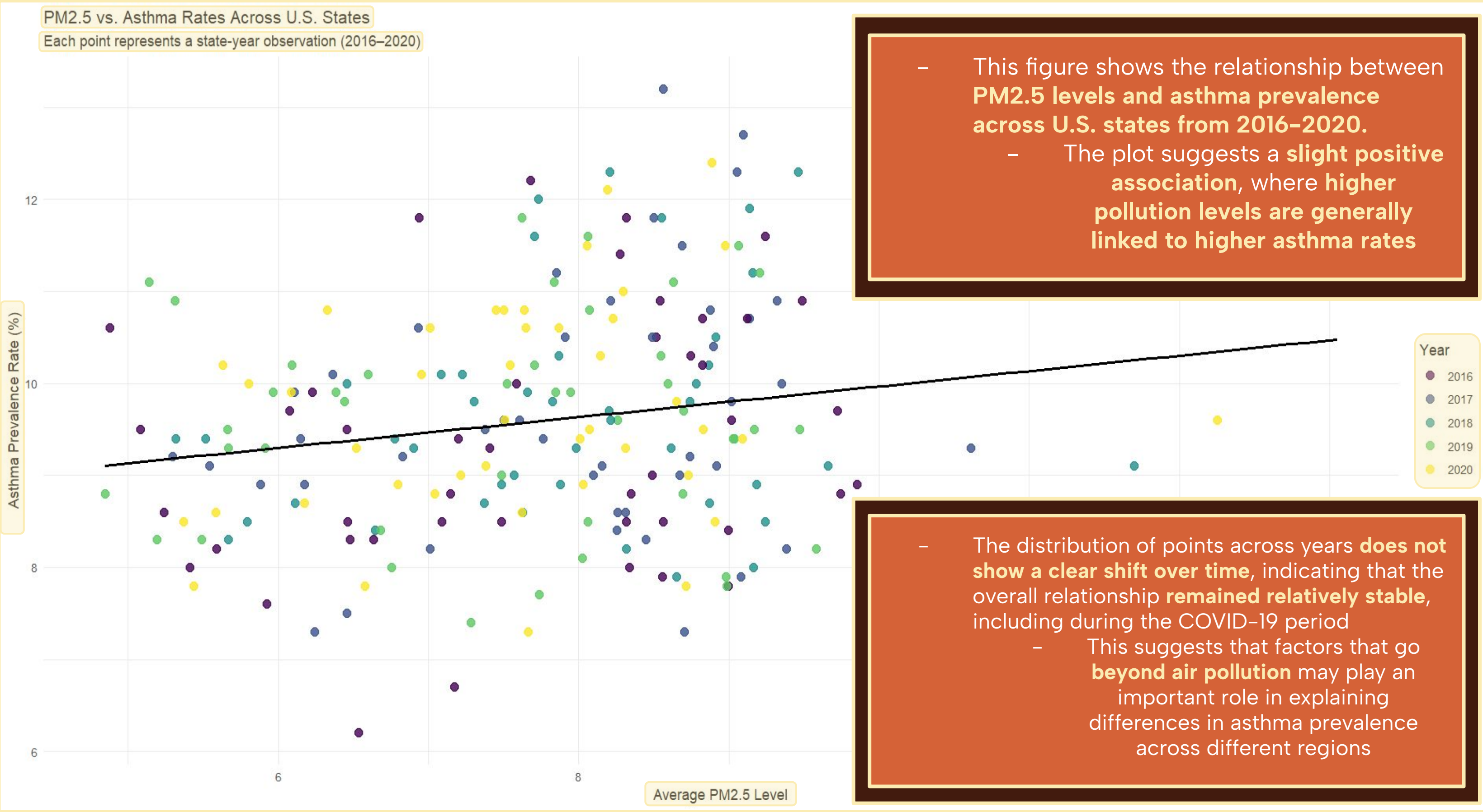
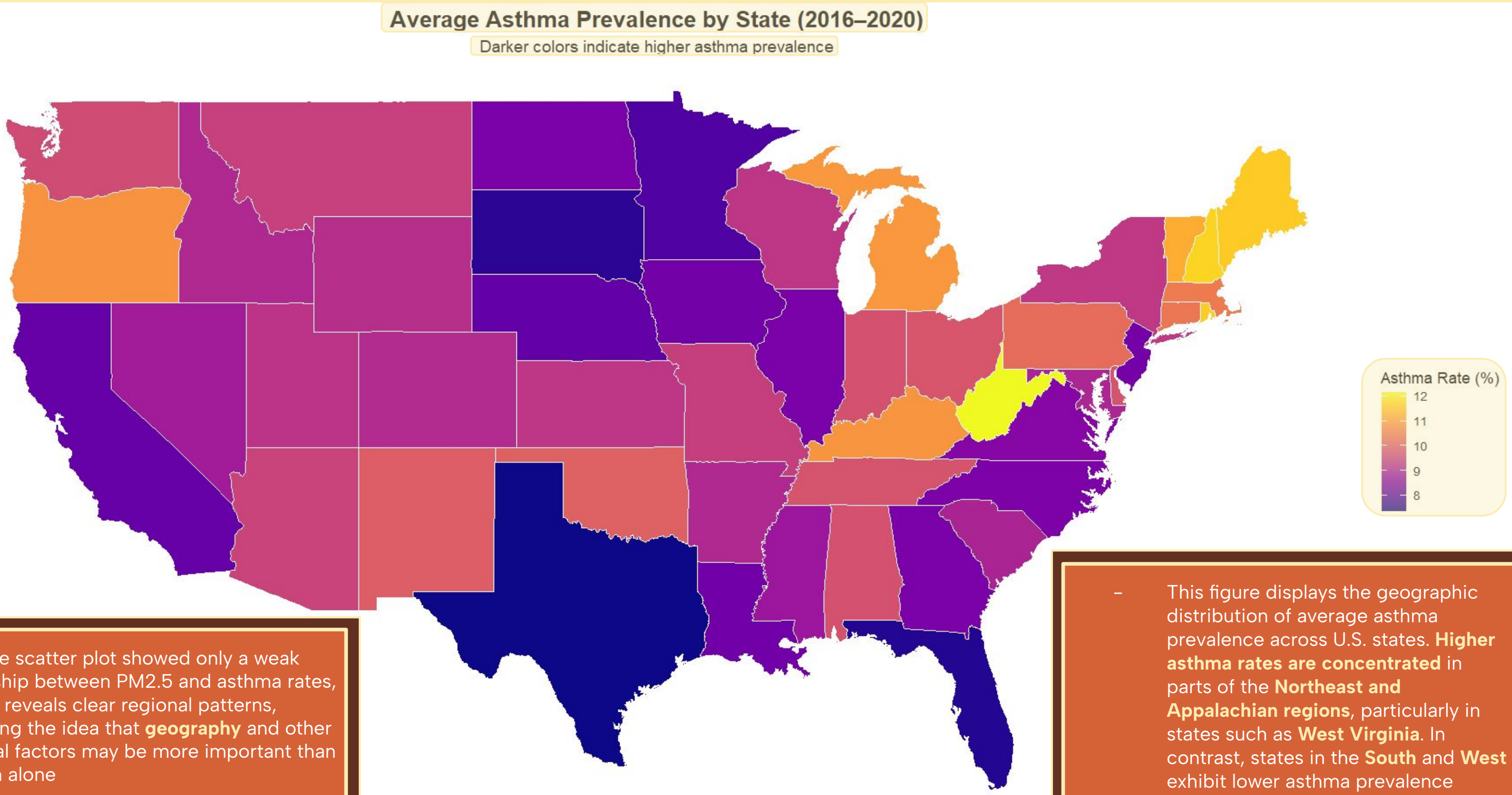
ABSTRACT

- The project examines the relationship between **PM2.5 air pollution** and **asthma prevalence** and **mortality** across **U.S. states** from **2016-2020**
    - **PM2.5 levels** measure **fine particulate matter** that is **less than 2.50 micrometers in diameter**, including **smoke**, **dust**, and **soot**
- It uses publicly available data from sources like the **EPA** and **CDC** to explore whether **higher pollution levels** are linked to **worse asthma outcomes**
- It also focuses on trends during the **COVID-19 pandemic** to identify **potential shifts in air quality and health outcomes**



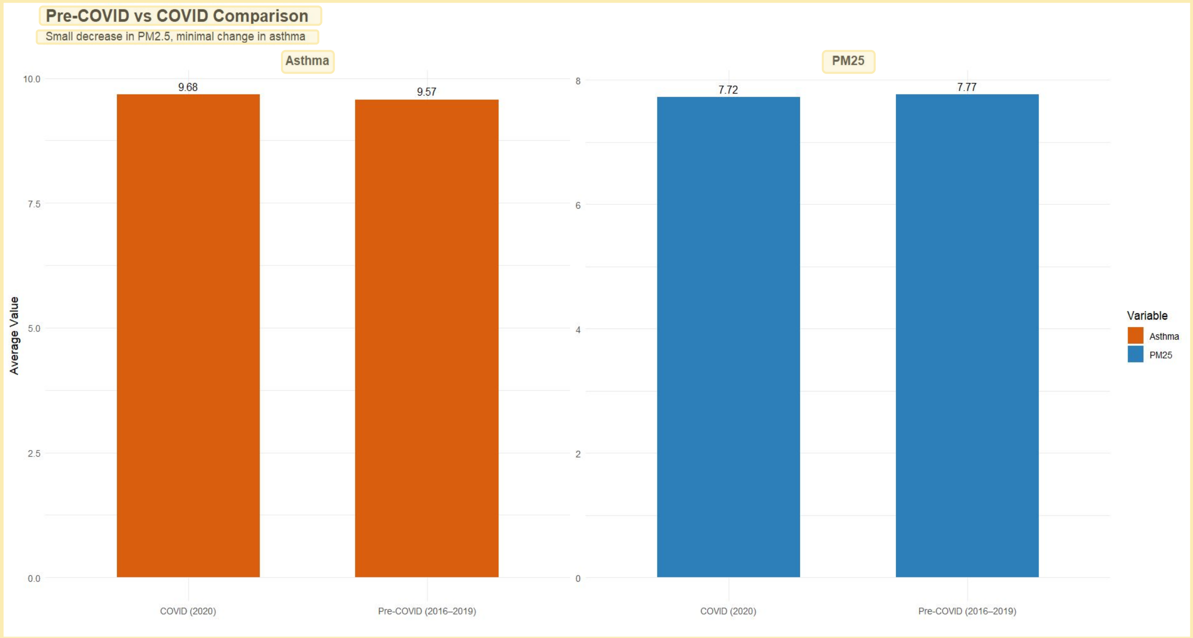
- This figure shows the relationship between **PM2.5 levels and asthma prevalence across U.S. states from 2016–2020.**
  - The plot suggests a **slight positive association**, where **higher pollution levels are generally linked to higher asthma rates**

- The distribution of points across years **does not show a clear shift over time**, indicating that the overall relationship **remained relatively stable**, including during the COVID-19 period
  - This suggests that factors that go **beyond air pollution** may play an important role in explaining differences in asthma prevalence across different regions



While the scatter plot showed only a weak relationship between PM2.5 and asthma rates, the map reveals clear regional patterns, reinforcing the idea that **geography** and other structural factors may be more important than pollution alone

- This figure displays the geographic distribution of average asthma prevalence across U.S. states. **Higher asthma rates are concentrated** in parts of the **Northeast and Appalachian regions**, particularly in states such as **West Virginia**. In contrast, states in the **South** and **West** exhibit lower asthma prevalence

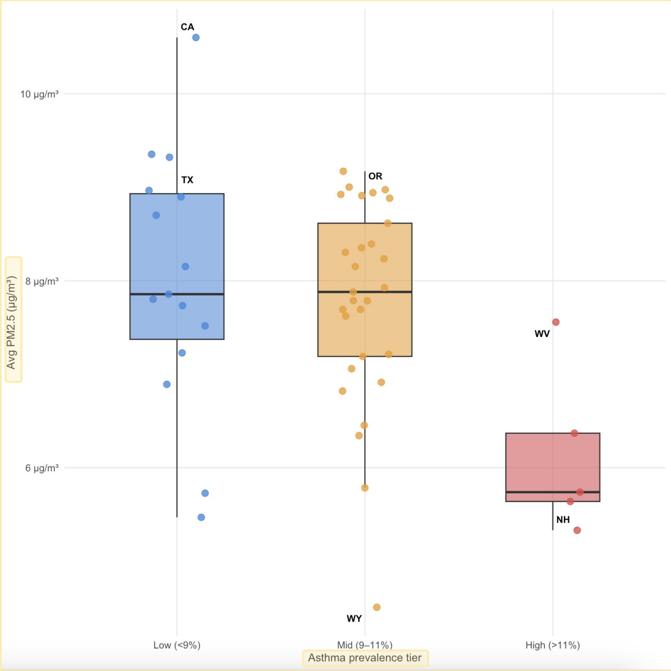


- This figure shows that while **PM2.5 levels decreased slightly during the COVID-19 pandemic, asthma prevalence increased marginally**
- Although the change is small, this lack of a corresponding decrease in asthma suggests that **reductions in air pollution do not immediately translate into improved asthma outcomes**
  - This reinforces our original idea that asthma prevalence is influenced by multiple factors **beyond short-term environmental conditions**

COMPARING STATES

- States with high asthma prevalence **do not consistently have the highest PM2.5 levels**, suggesting that **air pollution alone does not fully explain asthma burden**
- The distributions of PM2.5 across low, medium, and high asthma groups show substantial overlap, indicating only a **weak relationship between pollution levels and asthma prevalence**
- Several of the highest asthma states are concentrated in the **Northeast**, suggesting that regional factors, such as **climate**, **population density**, or **healthcare access**, may play an important role beyond air pollution

State	PM2.5	PM R	Asthma	AR	Tier
WV	7.56	31.0	12.14	1.0	High (>11%)
NH	5.33	48.0	11.74	2.0	High (>11%)
RI	6.37	41.0	11.64	3.0	High (>11%)
ME	5.74	44.0	11.62	4.0	High (>11%)
VT	5.64	46.0	11.28	5.0	High (>11%)
MI	8.39	15.0	11.00	6.0	Mid (9–11%)
KY	8.62	14.0	10.96	7.0	Mid (9–11%)
OR	9.00	5.0	10.94	8.0	Mid (9–11%)
MA	6.45	40.0	10.60	9.0	Mid (9–11%)
DC	8.35	16.0	10.54	10.0	Mid (9–11%)
CT	7.19	35.0	10.52	11.0	Mid (9–11%)
PA	8.88	12.0	10.36	12.0	Mid (9–11%)



Overall, while there is a slight positive association between PM2.5 and asthma rates, **the relationship is not strong, and significant variation remains unexplained**

CHALLENGES & FUTURE DIRECTION

Challenge #1

Inconsistent state datasets required extensive cleaning, merging, and alignment

Challenge #2

Weak relationships made meaningful interpretation and visualization design difficult

Direction #1

Examine confounding variables like income, healthcare, climate, and urbanization

Direction #2

Incorporate lagged analysis to capture delayed asthma responses